EMERYC CHUTTARSING

Quantitative Researcher | Machine Learning, Stochastic Processes & Financial Engineering

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EDUCATION

University of California, Berkeley, Berkeley, CA — Aug 2025 – May 2026

Master of Engineering in Industrial Engineering and Operations Research – FinTech Specialization

- Relevant coursework: Applied Stochastic Processes, Financial Engineering Systems, Applications of Machine Learning to Financial Markets, Optimization Analytics, and Data Science.

École Nationale Supérieure d'Arts et Métiers, Paris, France — Sept 2022 – June 2025

Diplôme d'Ingénieur (Master-equivalent) – Engineering, Mathematics, Statistics, and Optimization

Paris Dauphine University – PSL, Paris, France — Sept 2023 – June 2024

Final Year of Bachelor's Degree in Economics – Portfolio Management, Econometrics, Advanced Microeconomics, and International Economics (alongside Arts et Métiers)

PROFESSIONAL EXPERIENCE

EquitX – Batch-Auction Marketplace, Berkeley, CA — Jan 2026 – Present

Technical Co-Founder & Lead Engineer – joinequitx.com

- Designed agent-based demand simulations and willingness-to-pay models using Beta and Pareto distributions to study clearing-price dynamics and bidder outcomes.
- Built Python services and an OR-Tools CP-SAT matching engine for uniform-price allocation under quantity and adjacency constraints; operated a live real-money batch auction of approximately 100 tickets through settlement.

CCHE Entreprise, Geneva, Switzerland — Sept 2024 – Feb 2025

Assistant Project Manager Intern – Data Analytics & Predictive Modeling

- Built a CatBoost model from historical contractor execution records to predict realized completion times; the works-execution team used it to anticipate delays and support short-term planning.
- Coordinated execution reporting across 15 contractors and 100+ daily workers on a €200M development, translating field updates into schedule and budget decisions.

RELEVANT PROJECTS

Deep Reinforcement Learning for Cryptocurrency Markets — Sept 2025 – May 2026

UC Berkeley MEng IEOR FinTech Capstone – Team Quantitative Research Project

- Developed and evaluated Bitcoin allocation policies intended to remain robust across changing market regimes; compared D3QN, PPO, and SAC using 3-fold, 5-seed rolling walk-forward tests and selected models by worst-fold out-of-sample Sharpe.
- Used Rough Path Signatures as compact model features summarizing the order and shape of each 24-hour price path; achieved 91.4% versus 68.0% test accuracy in a controlled temporal experiment and reduced the measured generalization gap to 0.71 versus 1.39 for raw windows.
- Built a custom Gymnasium environment with cash, fractional Bitcoin holdings, total equity, fees, slippage, and exposure limits; replayed historical market dynamics while treating the strategy as a small price-taker.

Signature-Based Deep Hedging for Path-Dependent Options — Fall 2025

UC Berkeley – INDENG 223, Financial Engineering Systems II

- Investigated whether path-signature features could provide a more compact and effective representation of price histories for neural hedging of a path-dependent Asian option.
- Simulated price paths with geometric Brownian motion, incorporated transaction costs, and trained PyTorch policies with a CVaR objective; compared signatures with raw price histories, reducing model size by 24% and out-of-sample tail risk by 27%.

Multi-Stage Spotify Recommendation System — Fall 2025

UC Berkeley – INDENG 242B, Machine Learning Project

- Built a three-stage playlist-continuation system over 2.26M tracks and approximately 45M interactions; evaluation on 150K held-out playlists improved Recall@100 by 56% and NDCG@10 by 156% versus the retrieval baseline.
- ALS retrieved 1,000 candidates, a SASRec-style causal Transformer reranked 100 using sequence context, and MMR selected 20 final tracks while balancing relevance and diversity.

GPT-Style Decoder-Only Transformer — Fall 2025

UC Berkeley – INDENG 242B, Individual Machine Learning Project

- Built a language-modeling pipeline over approximately 40.7M tokens with BPE tokenization, causal multi-head attention, tied embeddings, and autoregressive generation; tracked validation loss and perplexity throughout training.
- Compared 3,000- and 5,000-token vocabularies and analyzed generation failures, finding improved token readability without overstating grammatical or long-range narrative coherence.

SKILLS

Programming: Python, SQL, R, MATLAB, VBA, LaTeX.

Libraries & Software: PyTorch, NumPy, Pandas, scikit-learn, CatBoost, XGBoost, Gymnasium, Gurobi, OR-Tools.

Languages: French (native), English (fluent).